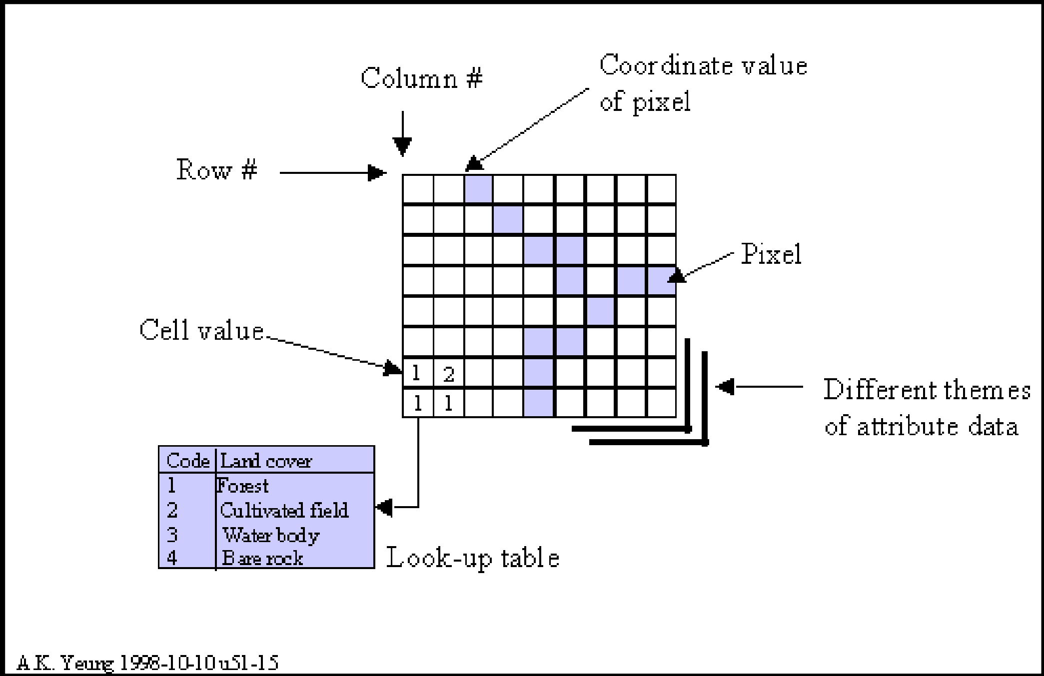
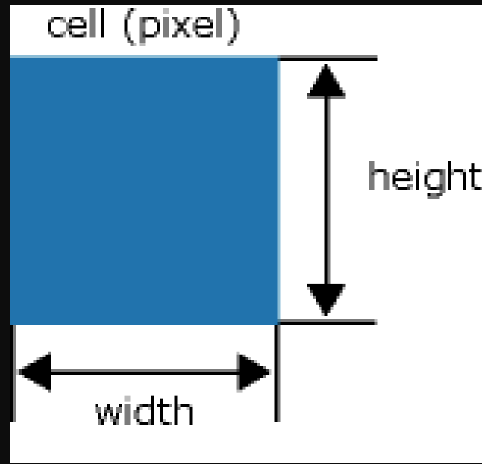
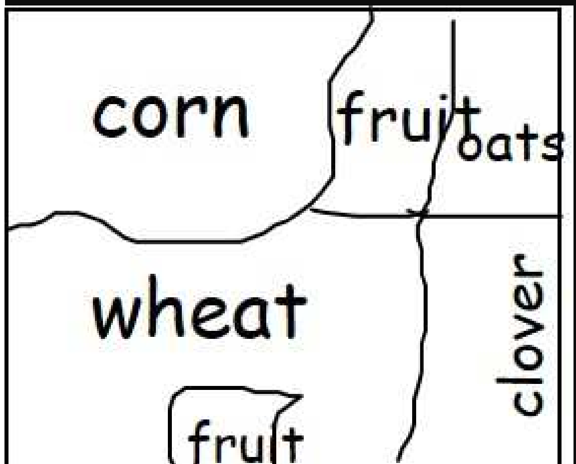
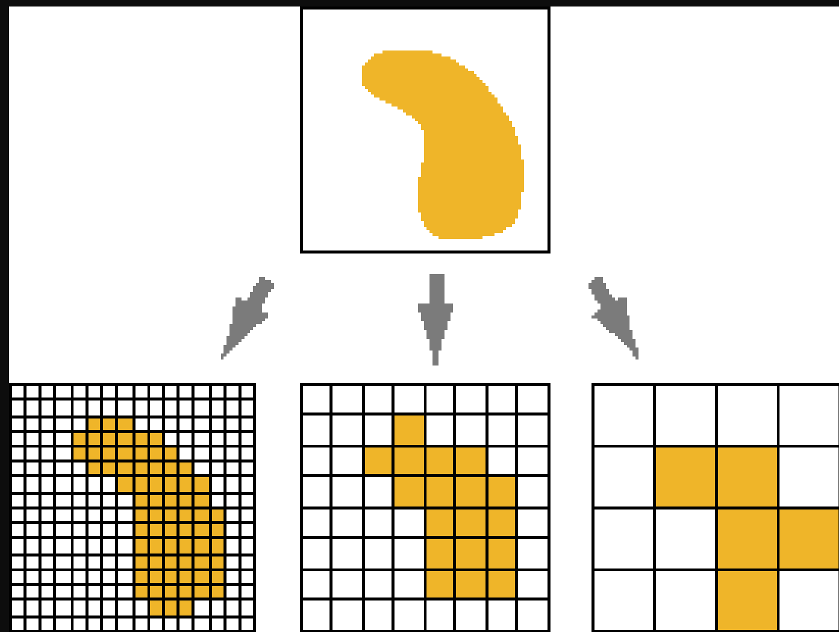
Raster Model







	0	1	2	3	4	5	6	7	8	9
0	1	1	1	1	1	4	4	5	5	5
1	1	1	1	1	1	4	4	5	5	5
2	1	1	1	1	1	4	4	5	5	5
3	1	1	1	1	1	4	4	5	5	5
4	1	1	1	1	1	4	4	5	5	5
5	2	2	2	2	2	2	2	3	3	3
6	2	2	2	2	2	2	2	3	3	3
7	2	2	2	2	2	2	2	3	3	3
8	2	2	4	4	2	2	2	3	3	3
9	2	2	4	4	2	2	2	3	3	3



Sample Data of a Raster

ASCII format

```
ncols         1000
nrows         900
xllcorner     76.6
yllcorner     12.5
cellsize      0.008333
NODATA_value  -9999
-9999 -9999 -9999 -9999 -9999 -9999 -9999 -9999 -9999 -9999 -9999
-9999 -9999 -9999 -9999 -9999 -9999 -9999 -9999 -9999 -9999 -9999
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```
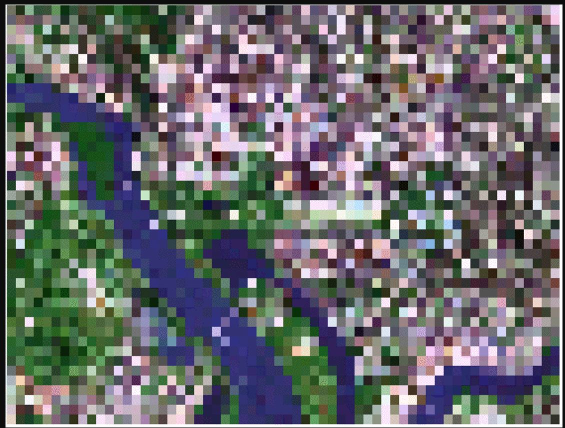
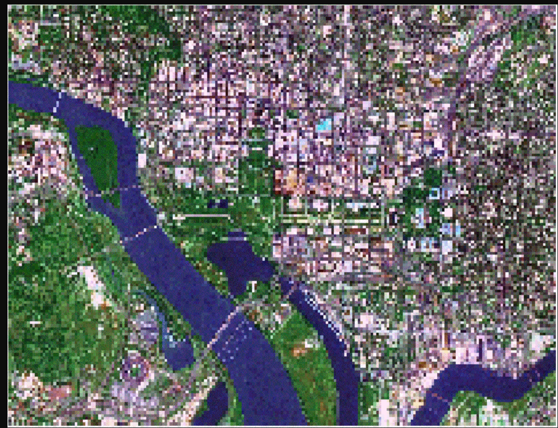

Parameter	Description	Requirements
NCOLS	Number of cell columns	Integer greater than 0.
NROWS	Number of cell rows	Integer greater than 0.
XLLCENTER or XLLCORNER	X-coordinate of the origin (by center or lower left corner of the cell)	Match with y-coordinate type.
YLLCENTER or YLLCORNER	Y-coordinate of the origin (by center or lower left corner of the cell)	Match with x-coordinate type.
CELLSIZE	Cell size	Greater than 0.
NODATA_VALUE	The input values to be NoData in the output raster	Optional. Default is -9999.

ASCII header information

Representing Data using Raster Model



100 meter



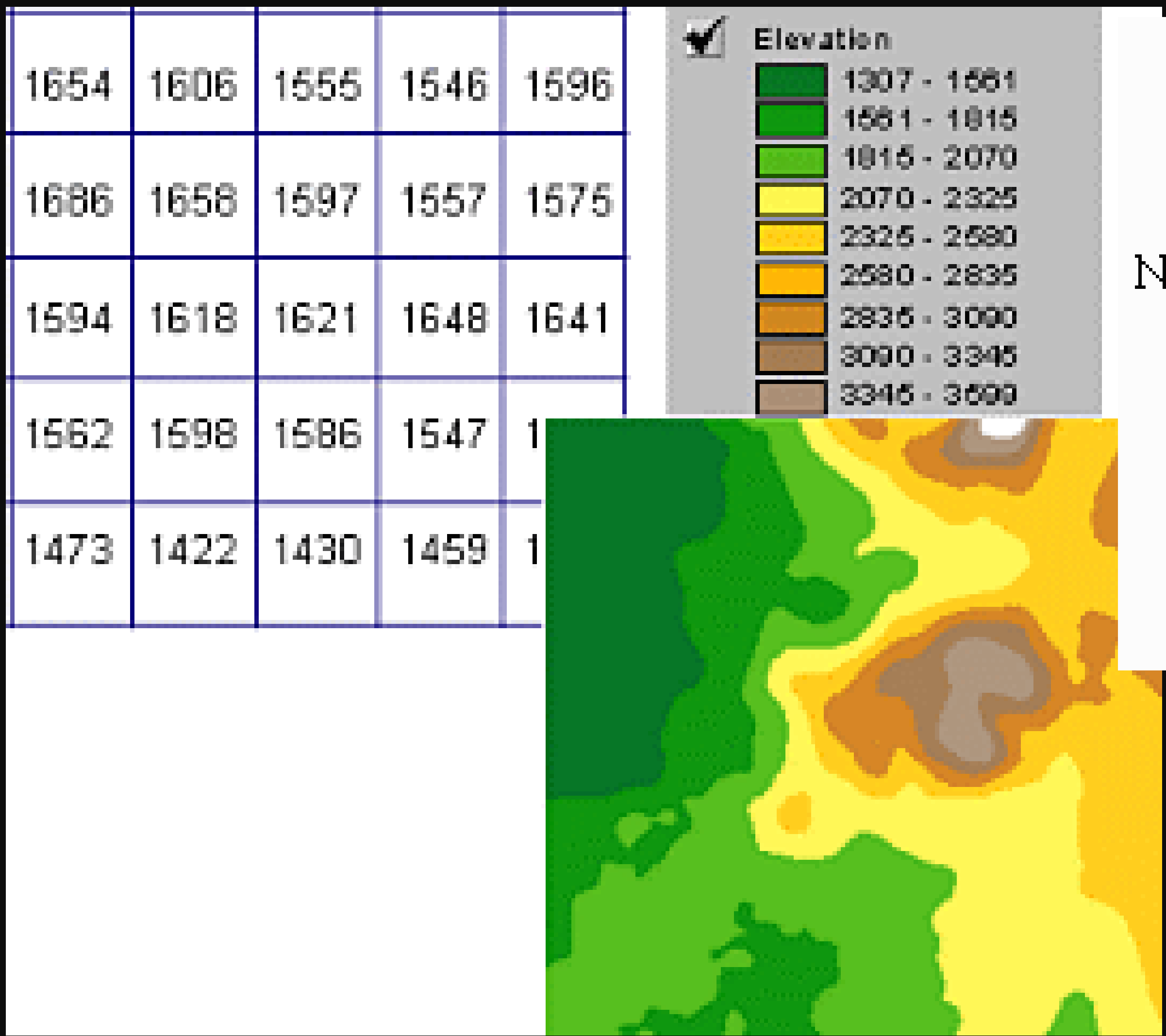
30 meter

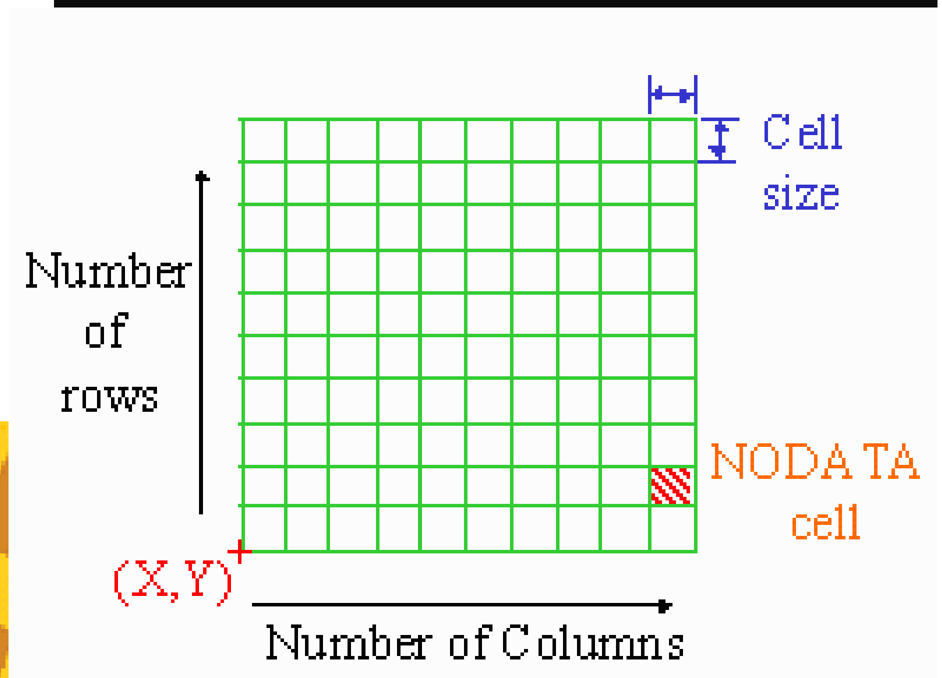
5 meter

<u>Raster data requires less processing than vector data</u>, but it consumes more computer storage space. Scanning remote sensors on satellites store data in raster format.

Raster data structures do not provide precise locational information therefore it may seem to be rather undesirable (DeMers, 1997).

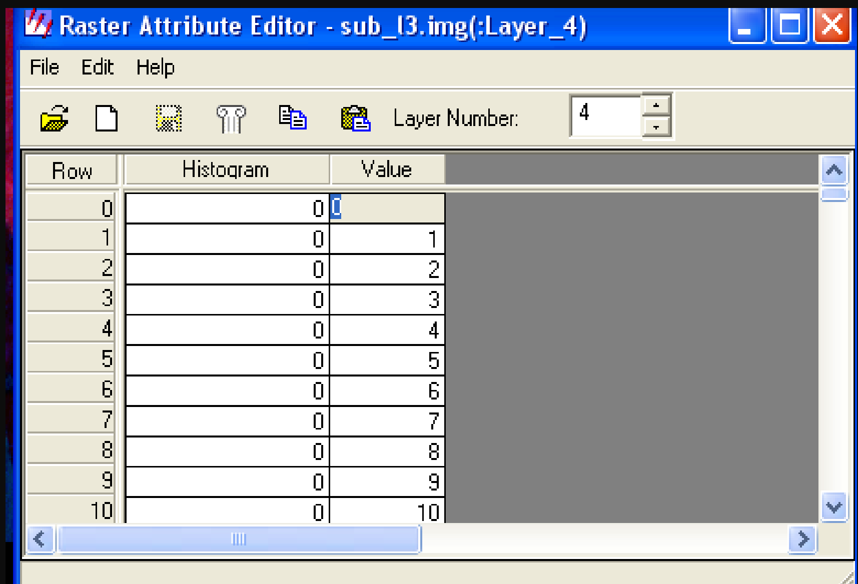
A **grid** defines geographic space as a matrix of identically-sized square cells. Each cell holds a numeric value that measures a geographic attribute (like elevation) for that unit of space.

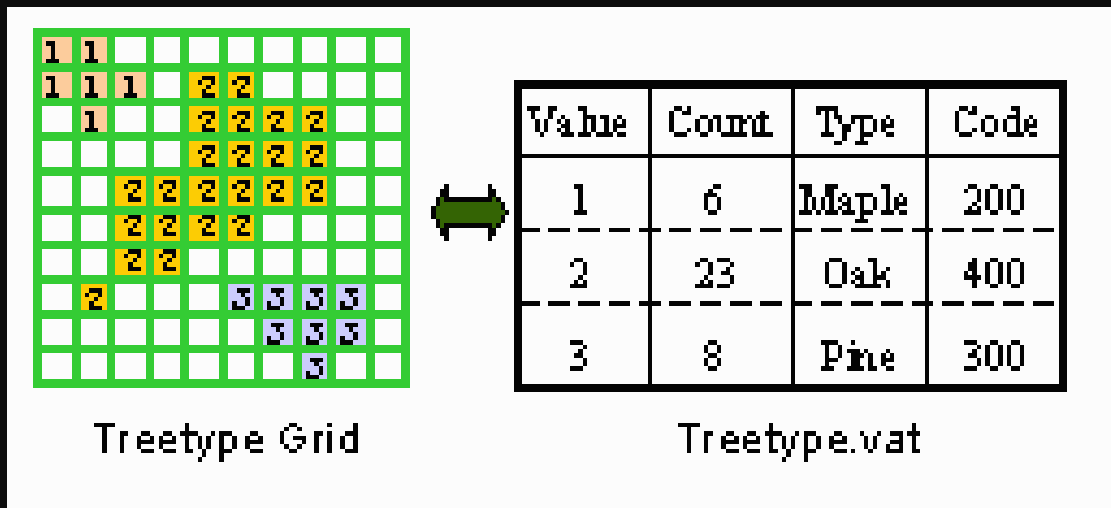




Raster characters

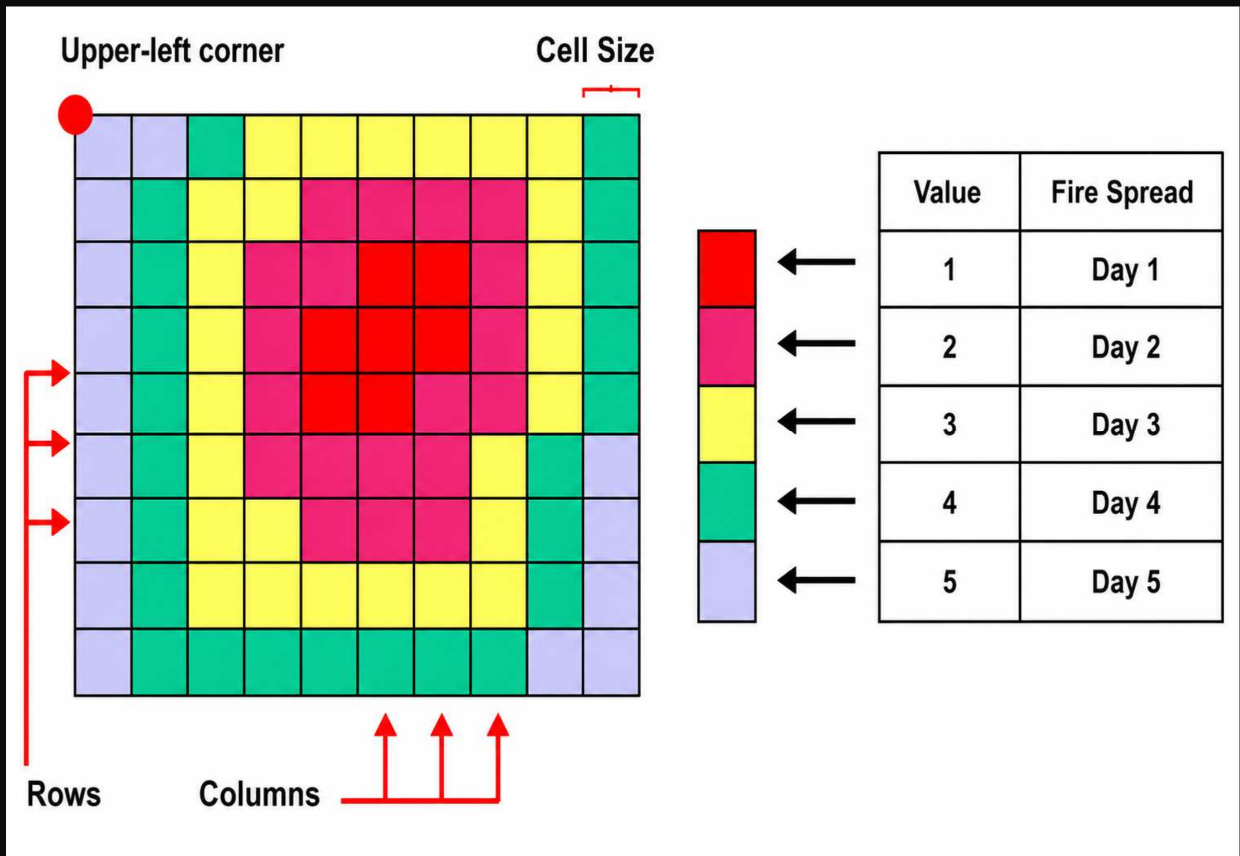
- <u>single</u> values associated with each cell
  typically 8 bits assigned to values therefore 256 possible values (0-255)
- easy to do overlays/analyses, just by 'combining' corresponding cell values
- simple data structure
  - directly store each layer as a single table (basically, each is analogous to a "spreadsheet")





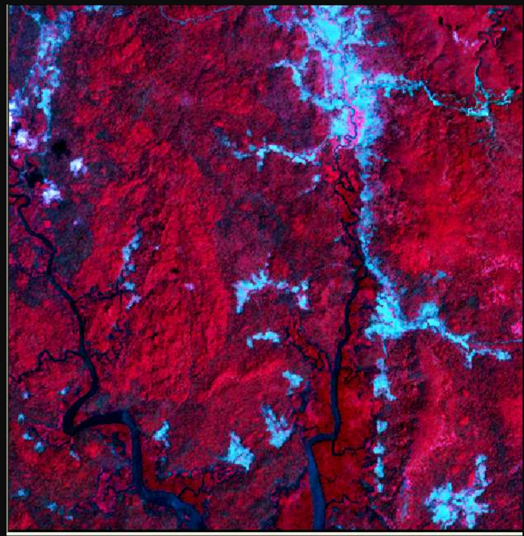
Because the raster data model is a regular grid, spatial relationships are implicit. Therefore, explicitly storing spatial relationships is not required as it is for the vector data model.

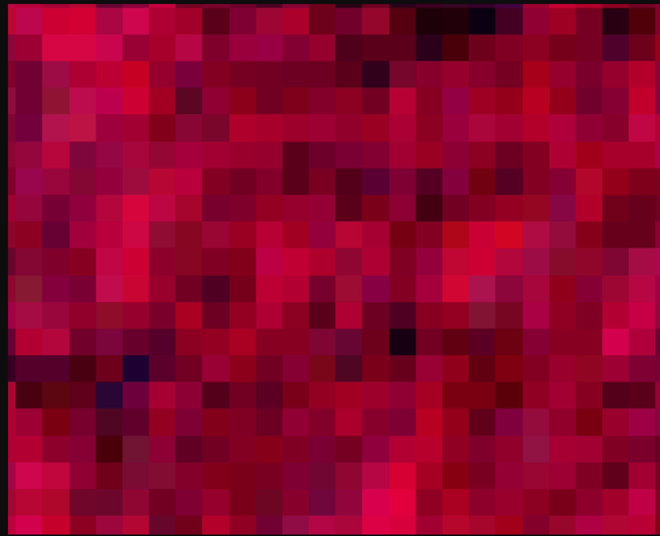
Adjacency and connectivity between cells can be derived from the data. e.g. find the minimum value around cell

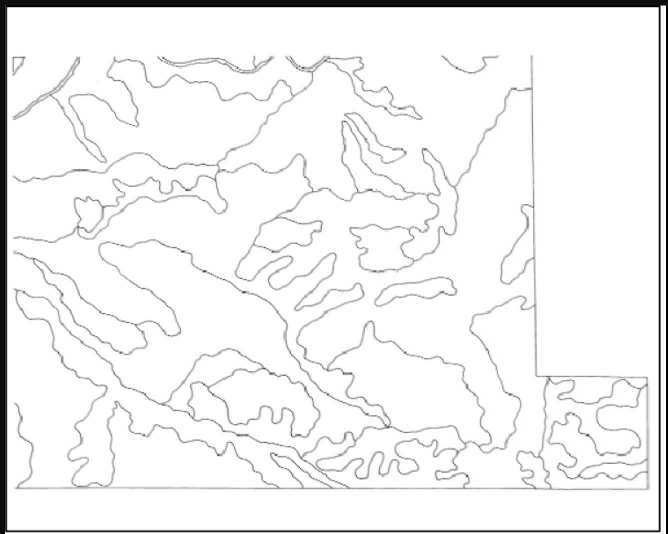


Raster Types

Satellite Imagery

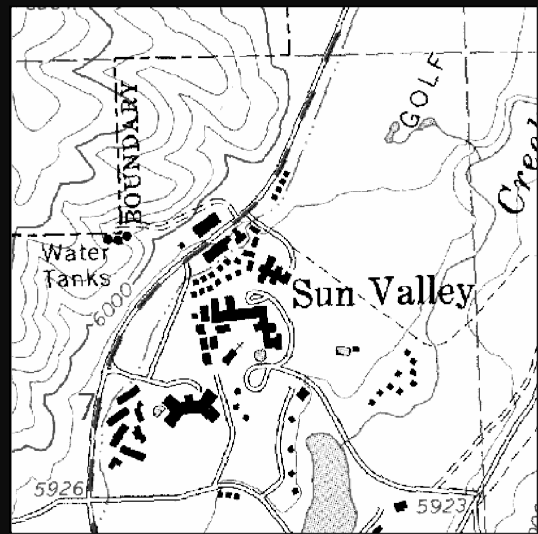






scanned file showing soil lines.

Digital Elevation Model

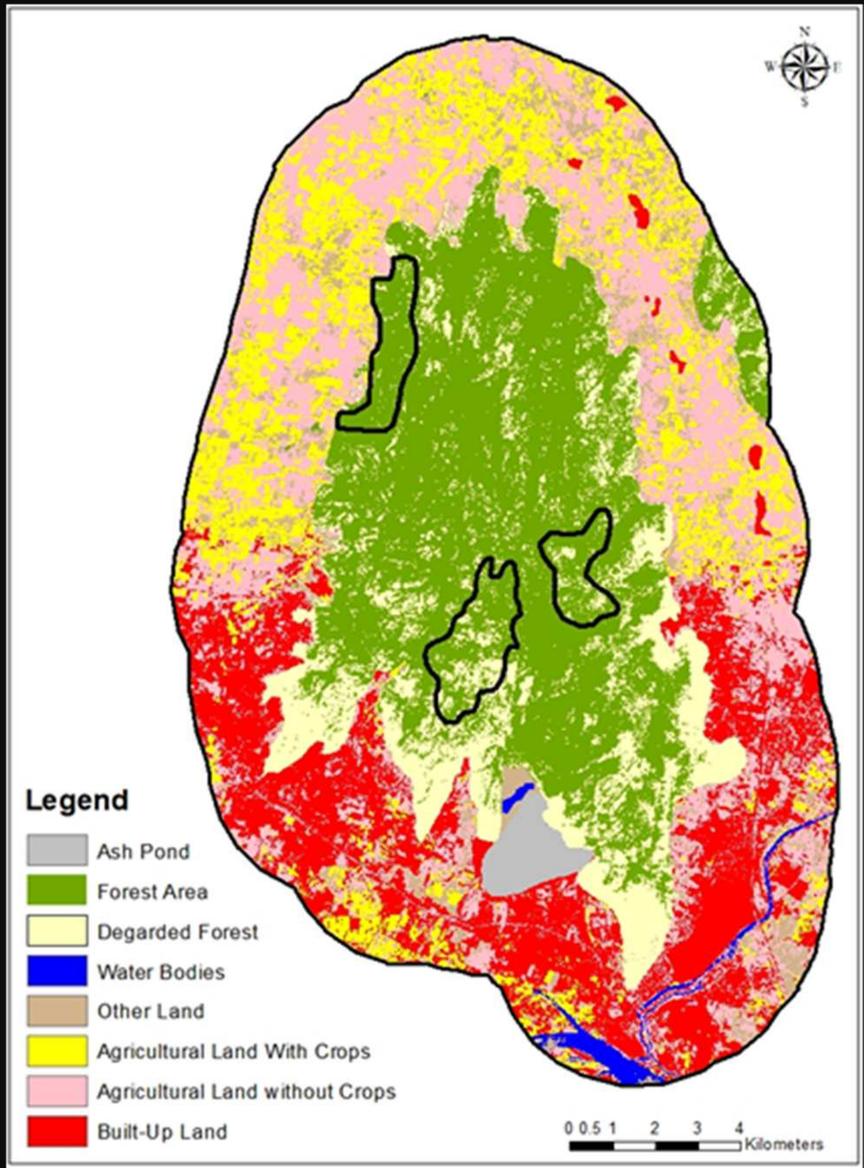


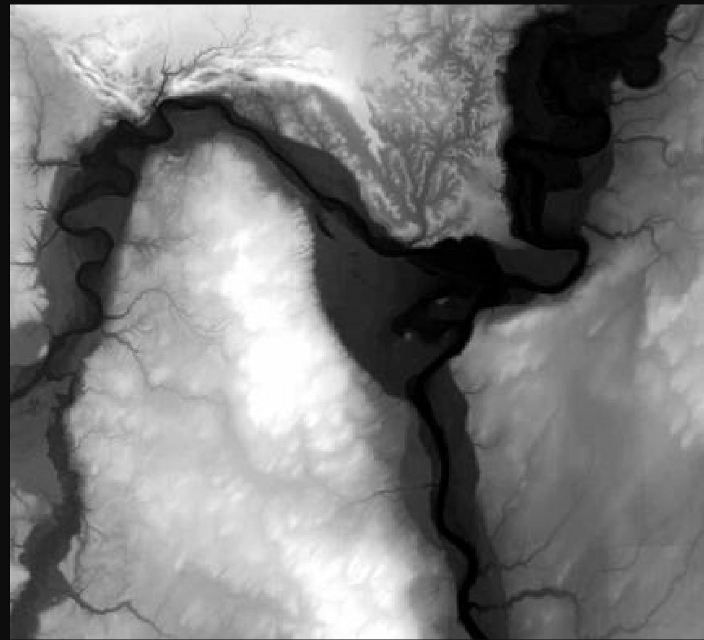
USGS DRG for Sun Valley

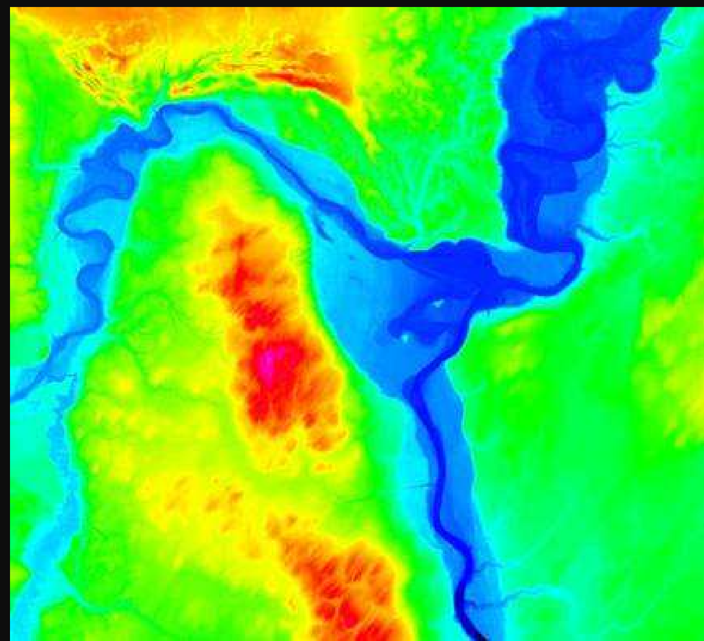
Attribute visualization

Elevation- Surface map

LULC – Thematic map

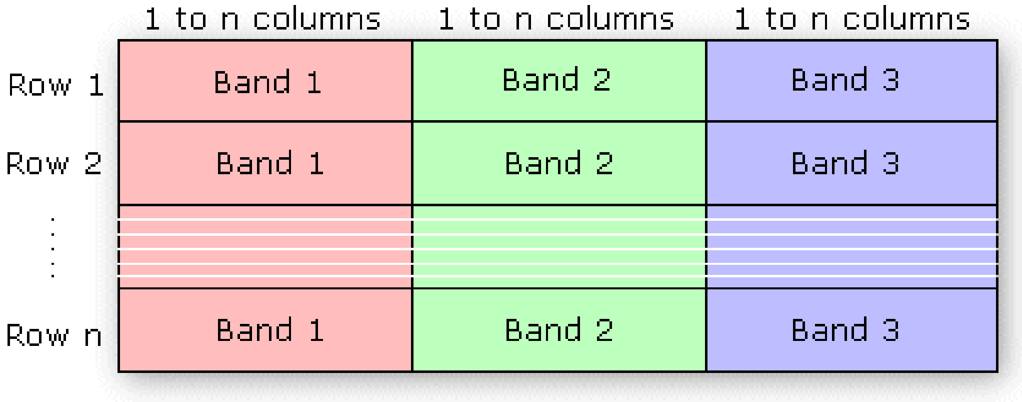




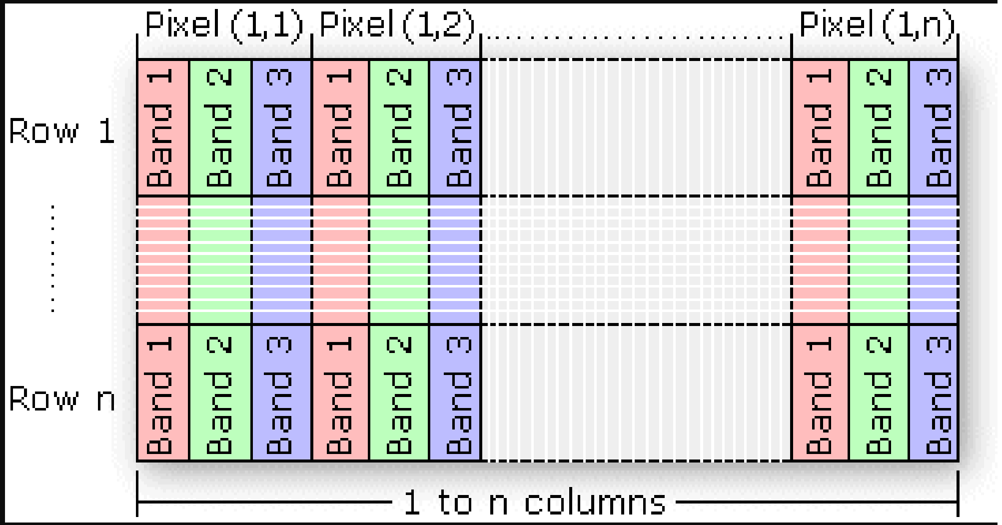


Raster Data Handling

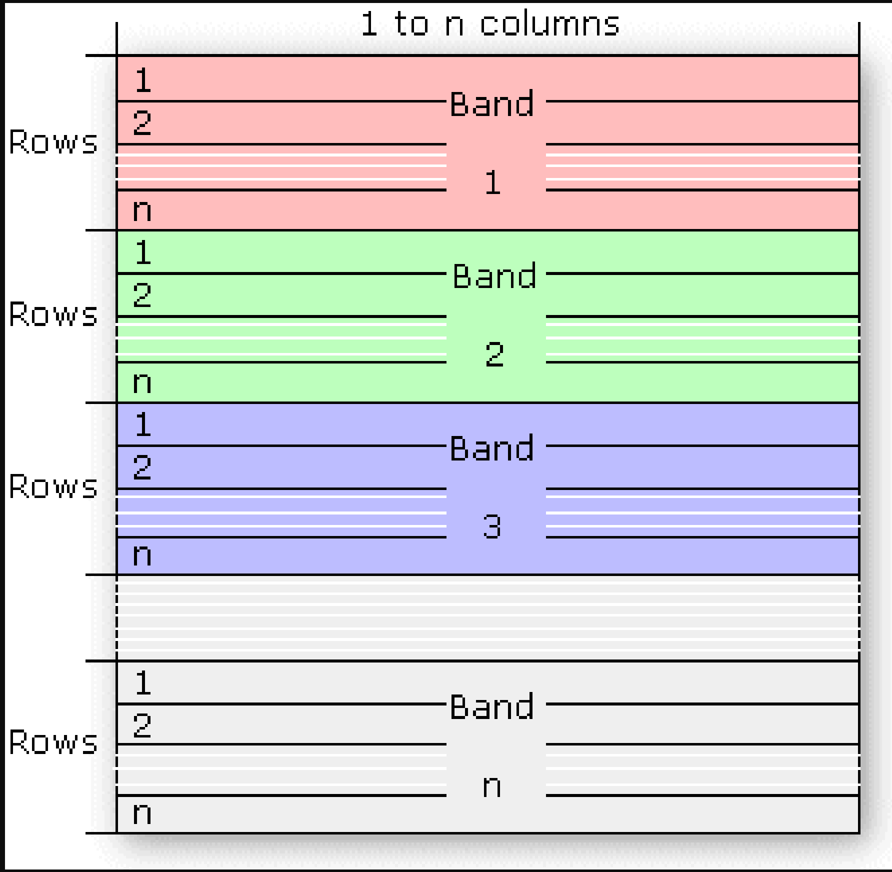
Raw data formats are



Band Interleaved by Line (BIL)



Band Interleaved by Pixel (BIL)



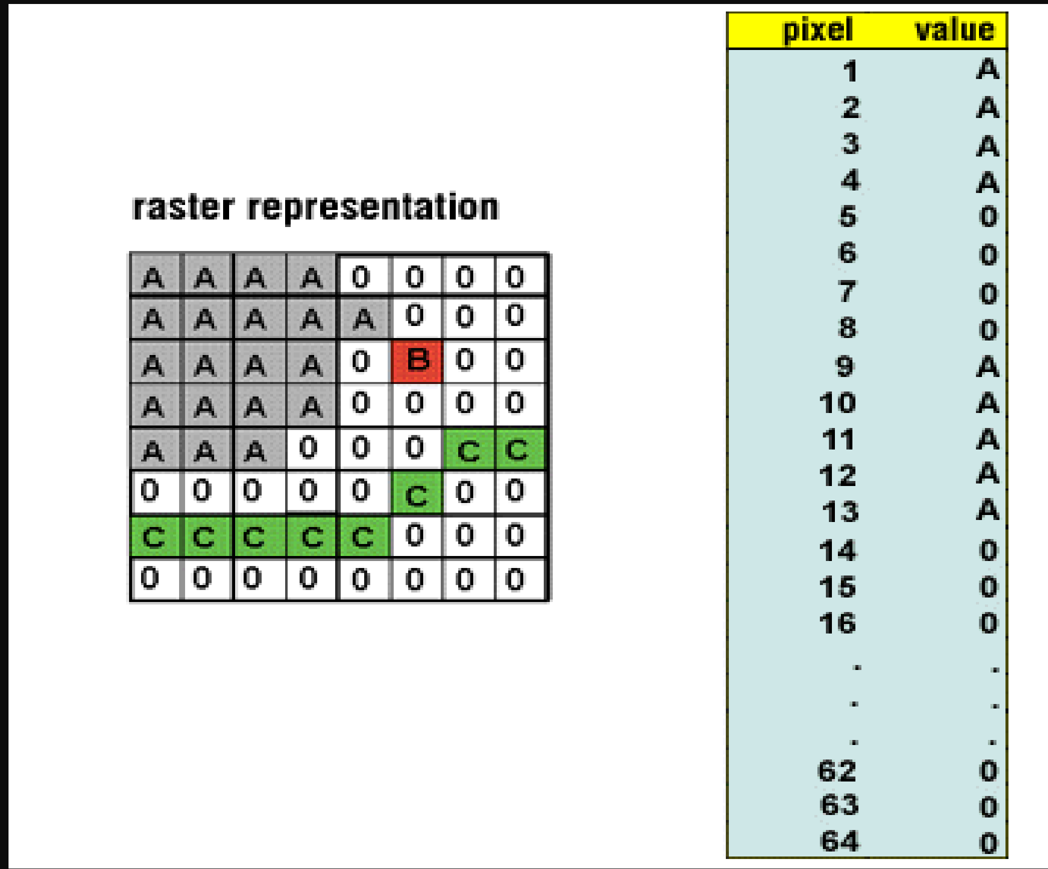
Band Sequential (BSQ)

Raster Data - Compression

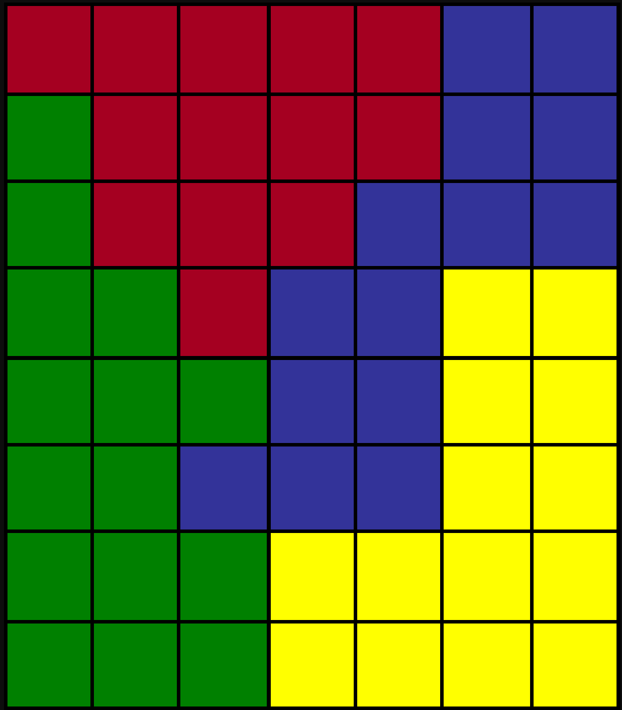
Conventional method / Cell by cell array

In this data structure, every pixel is given a single value, hence there is no compression when many like values are encountered.

Usually true for float-point continuous surface data



RUN-LENGTH CODING



A. Mixed Conifer

B. Douglas Fir

C. Oak Savannah

D. Grassland

Row-by-row coding:

CCCCCBB DCCCCBB DCCCBBB
DDCBBAA DDDDBAA DDBBBAA
DDDAAAA DDDAAAA

Run-length coding:

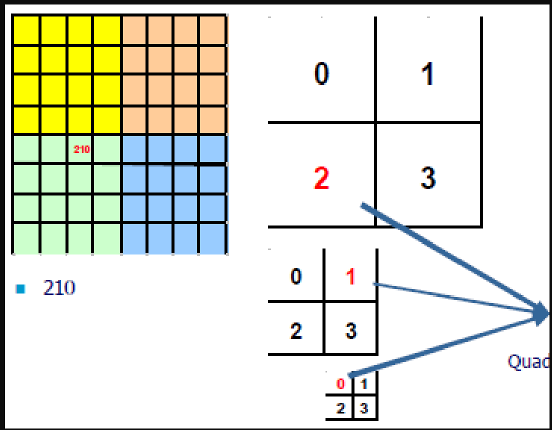
5C 2B 1D 4C 2B 1D 3C 3B 2D 1C 2B 2A 4D
1B 2A 2D 3B 2A 3D 4A 3D 4A

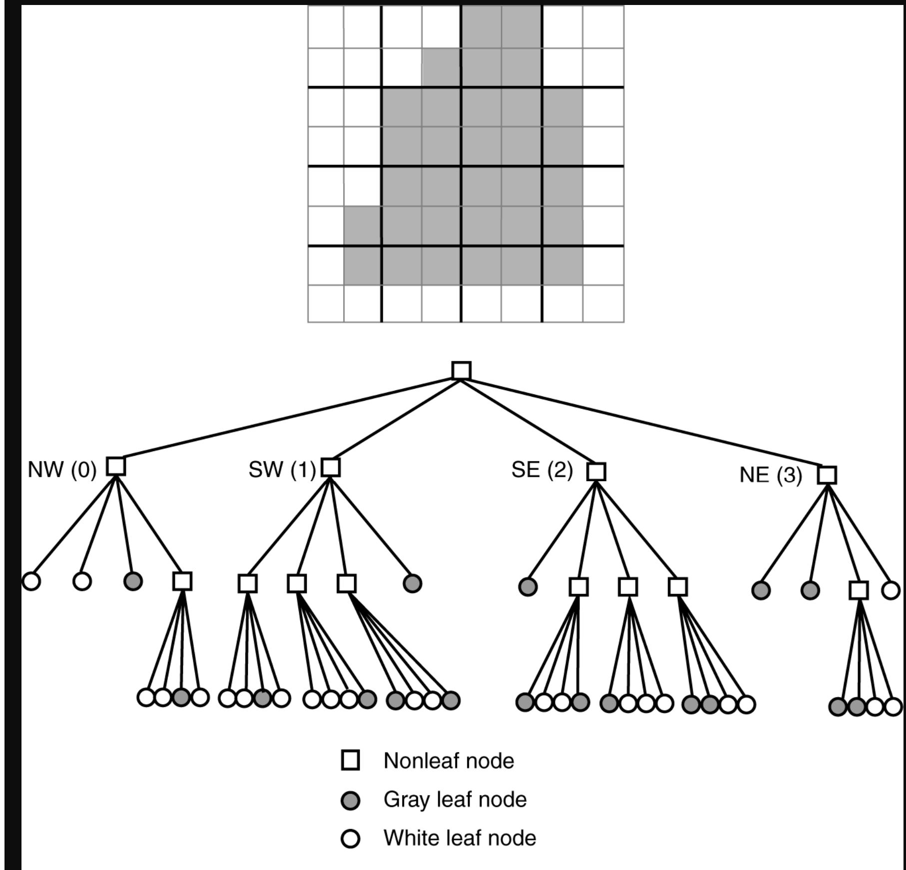
- **56 entries for 7x8 array, or**
- **22 pairs (44 entries) for 7x8 array**

Run-length encoding would be little use for DEM data or any other type of data where neighboring pixels almost have different values

lossless” compression

Quad Tree Representation





(02, 032), (102, 113, 120, 123, 13), (20, 210, 213, 220, 230, 231), (30, 31, 320, 321)

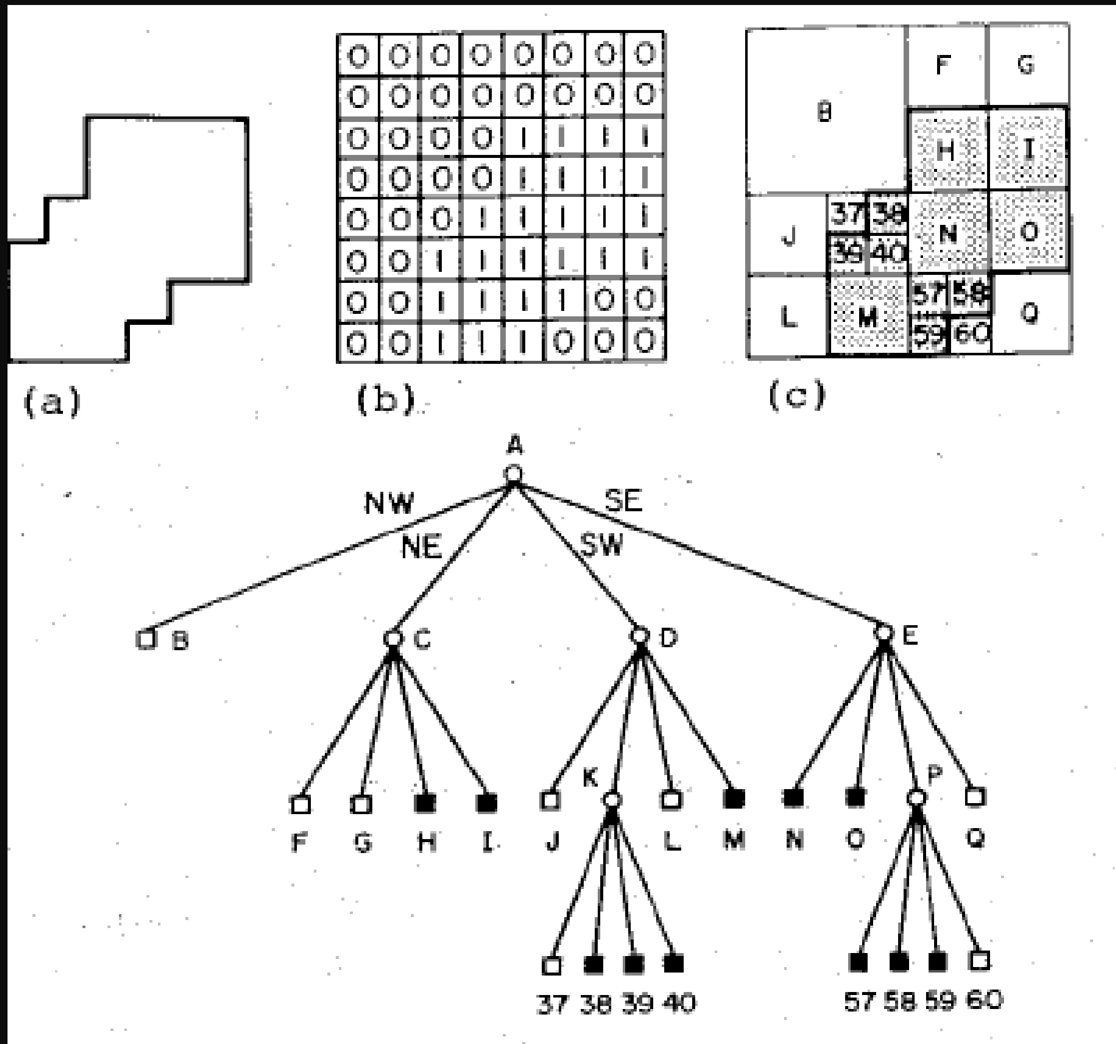
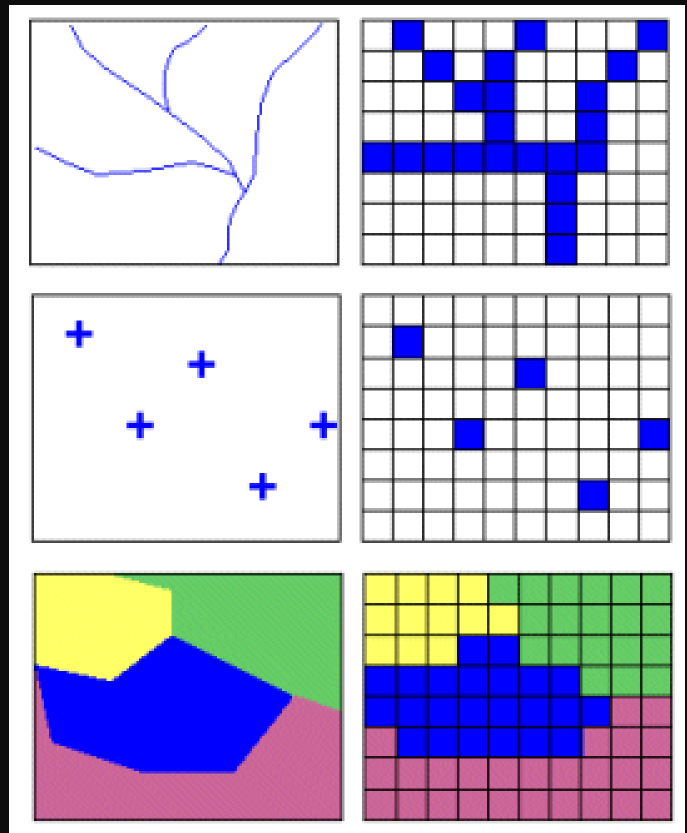
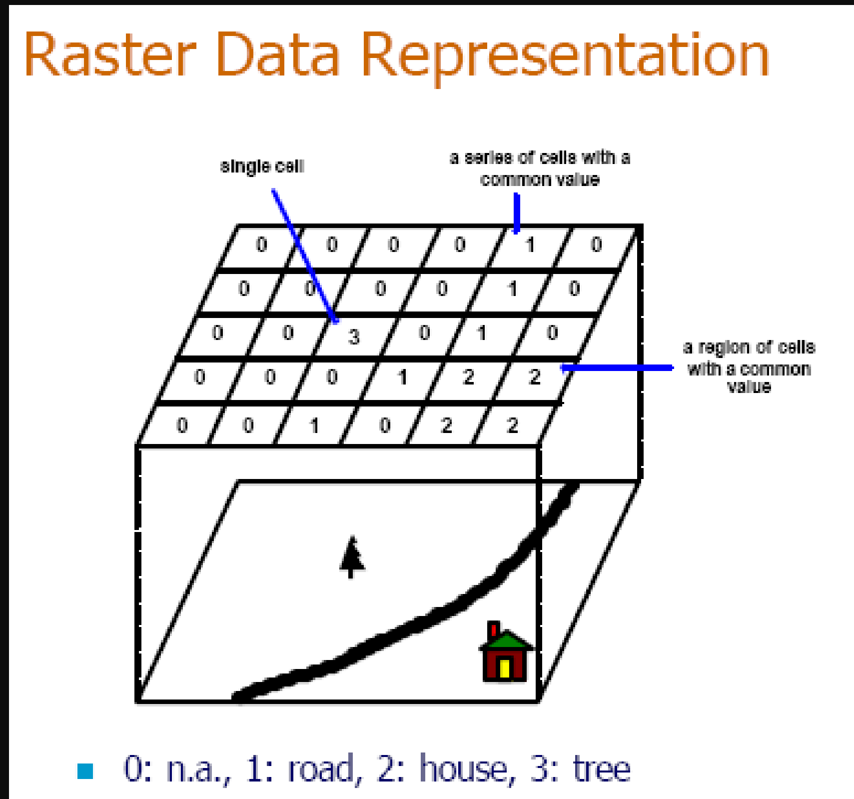


Figure 1. A region, its binary array, its maximal blocks, and the corresp quadtree. (a) Region. (b) Binary array. (c) Block decomposition of the region Blocks in the region are shaded. (d) Quadtree representation of the blocks in (c).

Vector and Raster Comparison

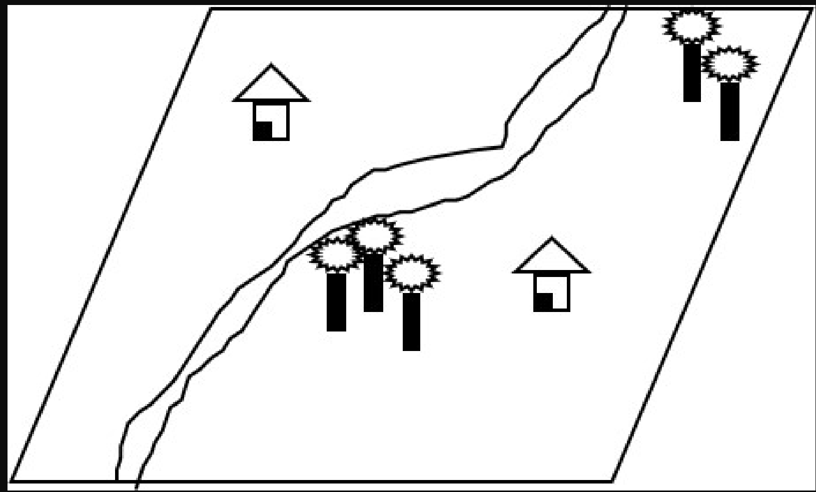
Raster is faster but vector is corrector" Joseph Berry





•A **point** feature is represented as a value in a single cell, a **linear** feature as a series of connected cells that portray length, and an **area** feature as a group of connected cells portraying shape.

Concept of Vector and Raster

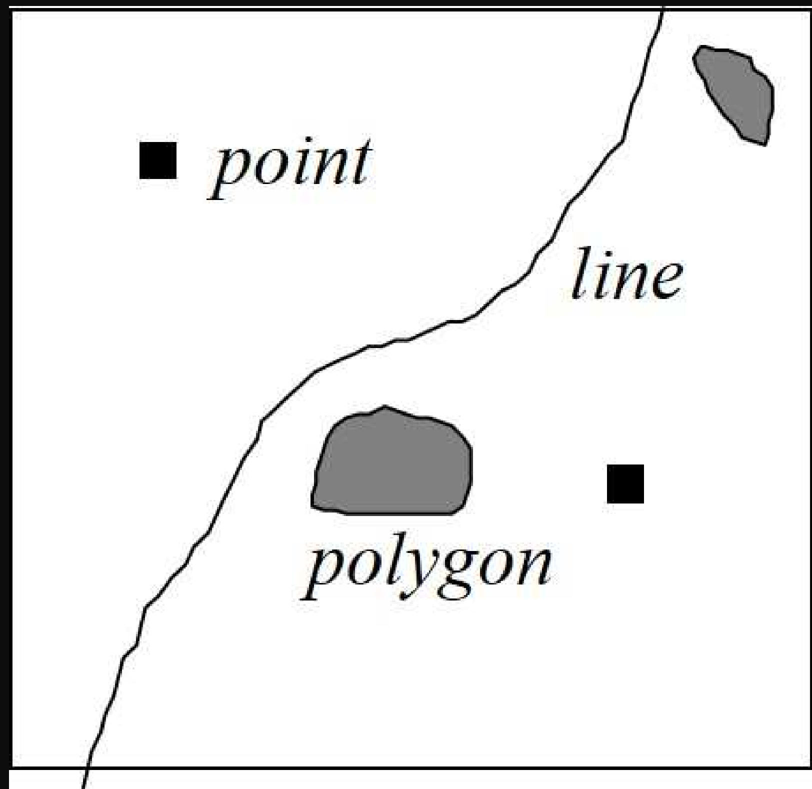


Real World

Raster Representation

	0	1	2	3	4	5	6	7	8	9
0								R	T	
1							R			T
2		H					R			
3							R			
4					R	R				
5				R						
6			R		T	T		H		
7			R		T	T				
8		R								
9		R								

Vector Representation



Vector vs Raster Graphics

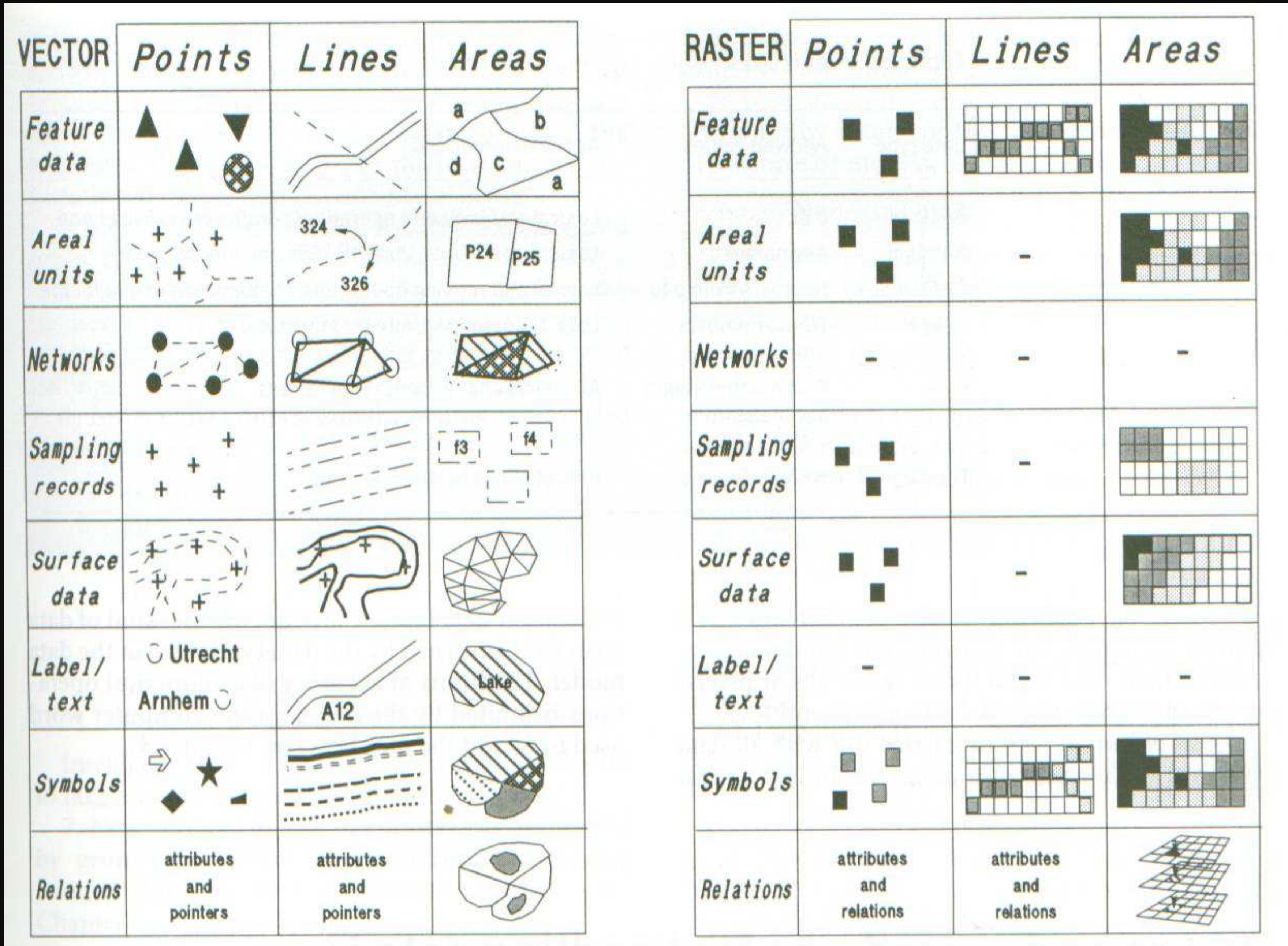


Image Source: Burrough, Peter A. and Rachael A. McDonnell. (1998). *Principles of Geographic Information Systems*. p 27.

Raster Data Model

Advantages

- Easy and fast computation.
- Well suited to quantitative modelling – Basic arithmetic operations
- Can integrate discrete and continuous data: Raster can represent both:

  Discrete data: distinct classes, such as forest = 1, agriculture = 2, water = 3.

  Continuous data: values that vary gradually, such as elevation or temperature.

Disadvantages

- The cell size determines the resolution at which the data is represented.
- Raster maps normally reflect only one attribute or characteristic for an area.
- Since most input data is in vector form, data must undergo vector-to-raster conversion.
- Most output maps from grid-cell systems do not conform to high-quality cartographic needs.

Vector Data Structures/Models

- Advantages
  - Good representation of entity data models
  - Compact data structure
  - Topology can be described explicitly – therefore good for **network analysis**
  - Accurate graphic representation at all scales
  - Retrieval, updating and generalization of graphics & attributes are possible
  - Graphic output is usually more aesthetically pleasing

Vector Data Model

The location of each vertex needs to be stored explicitly

For effective analysis, vector data must be converted into a topological structure. This is often processing intensive and usually requires extensive data cleaning.

Topology is static, and any updating or editing of the vector data requires re-building of the topology

Algorithms for manipulative and analysis functions are complex and may be processing intensive. Often, this inherently limits the functionality for large data sets, e.g.a large number of features.

Continuous data, such as elevation data, is not effectively represented in vector form. Usually substantial data generalization or interpolation is required for these data layers

Inter conversion of Data

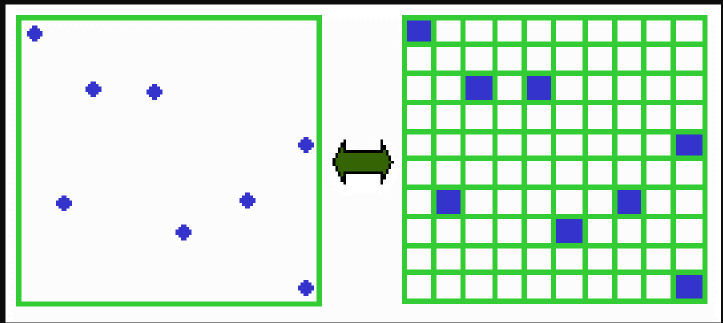
Often necessary to able to change from one basic data structure to another.

Loss of data.
Reduction in accuracy

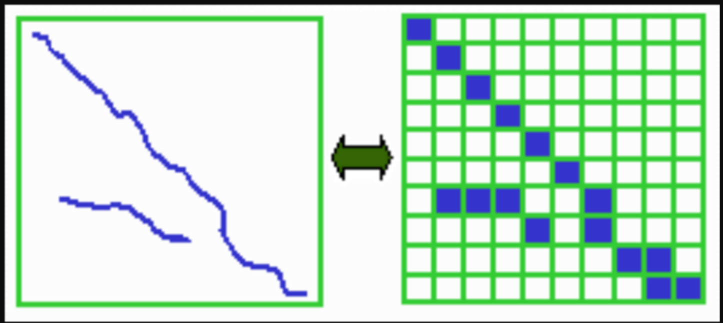
Rasterization (V2R)
Vectorization (R2V)

Rasterization is relatively easy.
Vectorization is much more complicated and difficult.

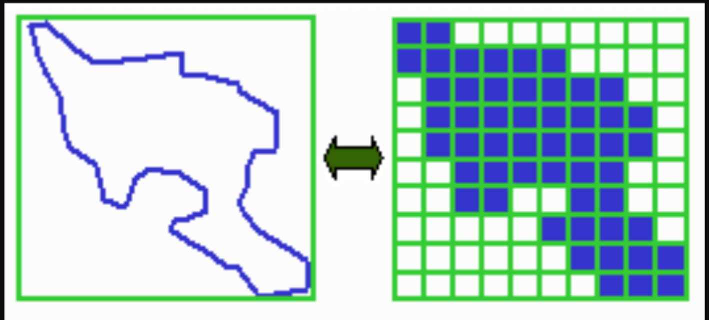
Data conversion from Vector to Raster vice versa



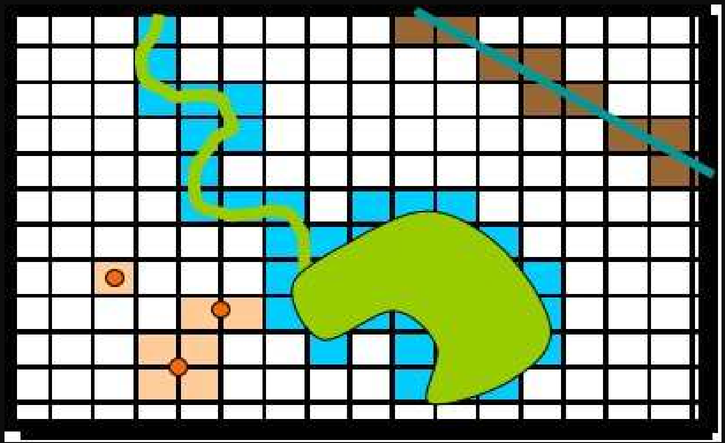
Points as Cells



Line as a Sequence of Cells



Polygon as a Zone of Cells



Vector to Raster

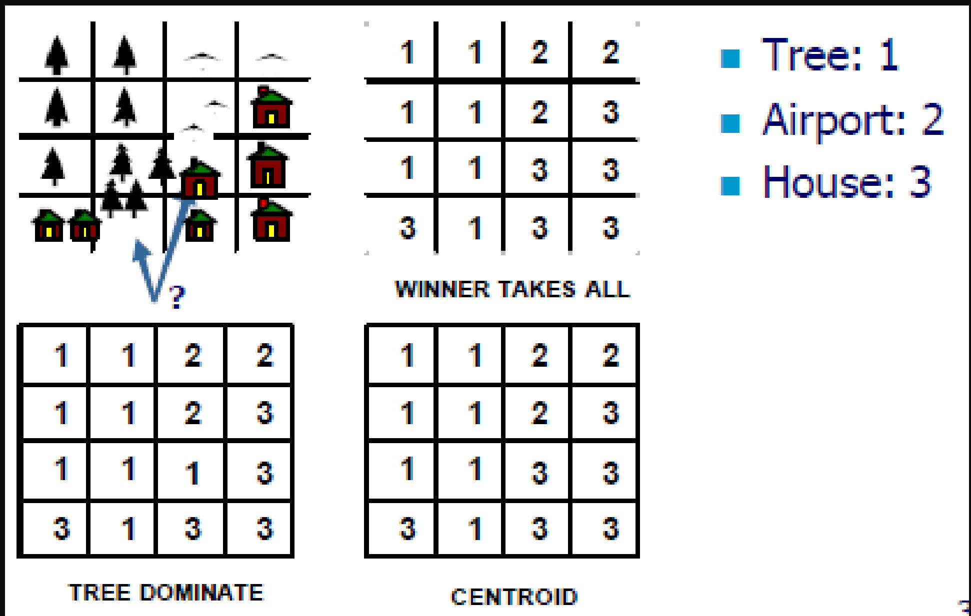
Raster Data Issue

precise location??

Jagged lines

Mixed pixel problem (V2R)

- Centroid method: Assigns to each cell the value of the feature that passes through the center of cell
- Winner takes all: major weighting
- Dominant: most important type.



MIXED PIXEL ISSUE

Water dominates

W	W	G
W	W	G
W	W	G

Winner takes all

W	G	G
W	W	G
W	G	G

Some Formats of Image datasets:

- Raw formats (BIL,BIP,BSQ)
- ESRI Grid datasets
- Windows bitmap images (BMP) [.bmp]
- Multiband (BSQ, BIL and BIP) and single band images [.bsq, .bil and .bip]
- ERDAS [.lan and .gis], IMAGINE [.img]
- IMPELL Bitmaps [.rlc]
- JPEG [.jpg]
- MrSID [.sid]
- National Image Transfer Format (NITF)
- Sun rasterfiles [.rs, .ras and .sun]
- Tag Image File Format (TIFF) [.tiff, .tif and .tff]

Common Vector Data Formats

- Shapefile (.shp, .shx, .dbf) – traditional and widely supported
- GeoJSON (.geojson / .json) – web-based, lightweight format
- GeoPackage (.gpkg) – modern, open, SQLite-based format
- KML/KMZ (.kml / .kmz) – commonly used with Google Earth
- File Geodatabase (.gdb) – Esri geodatabase format
- GeoDatabase / Enterprise Geodatabase – spatial database storage
- MapInfo TAB (.tab) – MapInfo vector format
- GML (.gml) – XML-based geographic data format
- GPX (.gpx) – GPS tracks, routes and waypoints
- DXF/DWG (.dxf / .dwg) – CAD-based formats often used for spatial data